

SmartDrive: Semi-Autonomous Scaled Robotic Car

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Term Project

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Abstract

This project implements semi-autonomous behavior on a scaled robotic car by fusing vision, speed feedback, and distance sensing. A Pixy2 camera classifies traffic light states (red, yellow, green), a tachometer provides real-time speed, and a distance sensor enforces a 6-inch buffer when following a taught object. Yellow light decisions use time-to-intersection (distance over speed) to choose whether to stop or proceed. Motor commands are smoothly modulated via PWM for stopping, cruising, and following. Key results include reliable red light stopping, stable speed maintenance on green, correct borderline yellow decisions, and responsive object following with collision avoidance. The system is modular and ready for future enhancements.

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Introduction

Autonomous and semi-autonomous driving systems reduce human error and improve safety. In this project, we build a scaled robotic car that perceives its environment using vision and sensing, then makes context-aware drive decisions. The core features implemented are traffic light state recognition, maintaining safe following distances to a taught object, and speed-aware decision making under ambiguous yellow light conditions. This report describes the design, implementation, testing, and team workflow across three in progress reviews leading to the final system.

Approach

1. Scientific & Engineering Principles

Computer Vision and Classification: The Pixy2 camera is trained to recognize traffic light colors (red, green, yellow) and a separate object for following. Color signature tuning and ambient light adjustment ensure robustness.

Control Systems: Motor speed is modulated via PWM to achieve smooth acceleration and deceleration. Feedback from the tachometer allows velocity-aware decisions.

Decision Logic: State machine governs behavior based on traffic light input:

- Red: smooth deceleration to stop within 2 to 6 inches of the light.
- Green: maintain target speed.
- Yellow: compute time to intersection ($\text{distance} / \text{current speed}$) and compare with yellow duration to decide stop versus proceed.

Collision Avoidance: While following a taught object, a minimum buffer of 6 inches is maintained using real-time distance sensing; if the object slows or stops, the car adjusts to prevent contact.

2. - Tools & Technologies

Hardware	Software
Pixy2 Camera w/ Mounting Bracket	Arduino IDE
Traffic Light Prototype (LEDs)	Pixy2 Library for Signature Detection
Tachometer & Distance Sensor	Serial Logging
DC Motors w/ Driver	
Arduino Microcontroller	

3. - Systems Integration

The system fuses vision, speed, and proximity data into a unified control pipeline. The high level architecture is:

- **Input layer:**
 - Pixy2 delivers classified traffic light state and object position
 - The tachometer provides current speed
 - The distance sensor gives a gap to the object or the light
- **Decision logic layer:**
 - Traffic light state machine evaluates red, green, and yellow conditions
 - Yellow logic computes $\text{time_to_intersection} = \text{distance}/\text{speed}$ and compares it to the preset yellow duration to choose stop or go
 - Object following logic monitors the buffer gap and overrides the speed if the separation violates the safety threshold.
- **Actuation layer:**

- Motor PWM is adjusted smoothly based on the desired velocity or braking command
- Composite behavior emerges when multiple inputs (e.g., green light while following an object) are reconciled (priority rules defined in code)

Data flows in real time: vision and sensors feed into the decision module, which outputs target speed commands. Feedback from the tachometer allows closed-loop correction, reducing drift and improving responsiveness, especially critical in yellow light borderline cases.

4. - Teamwork To Engineer the Concept

For the project we were able to collaborate as a team to build the car in the team kickoff. The IPR1 the team needed to print a bracket and build a stop lamp. The bracket was 3D printed by David, the stop lamp was wired and the stop lamp post by Eric. IPR 2 was programmed with the Pixy camera to follow an Orange basketball. This was coded and demoed by Dale and Sandra. Additionally for IPR 2 the Pixy was programmed to recognize each light as a specific signature; this was performed by Eric, George, David in the end, although Dale and Sandra were also part of the initial assembly. For IPR 3 the LM393 speed sensor was installed by Eric. Part #2 for IPR 3 was a group effort with all 5 teammates bouncing ideas to get the robot to function properly. The team met for the Extra credit for the project. Sandra and David started the document and power point presentation aside from our meeting dates, with the rest of the team helping out.

Milestones & Schedule

DATE	TASK
July 2nd, 2025	Meet To Assemble the Car
July 5th, 2025	Meet on Discord before meeting
July 5th, 2025	Kickoff & IPR 1
July 12th, 2025	Discuss IPR II After Class
July 15th, 2025	Work on IPR II
July 29th, 2025	Work on IPR II
July 31st, 2025	Complete ½ of IPR II
August 1st, 2025	Finished all of IPR II and recorded
August 2nd, 2025	Present IPR II
August 5th, 2025	Start on IPR III
August 6th, 2025	Complete IPR III
August 9th, 2025	Present IPR III
August 13th, 2025	Worked on extra credit
August 16th, 2025	Submit everything

Budget

- Pixy2: Gifted by professor (otherwise ~\$70)
- LM393 speed module: Gifted by professor (otherwise ~\$2)
- Chassis + motors: \$58
- Misc Wires: \$15
- Total: ~ \$73